

Part 1

Suppose we had a linear regression model with one feature that learned the following bias and weights:

$$b=1 \quad w_1=3$$

Further, suppose the following feature values for a given example:

$$x_1=0$$

Compute the linear regression prediction for this particular example using python in Google Colab. Make sure to plot the solution. Provide an analysis for the output.

```
import matplotlib.pyplot as plt
import numpy as np

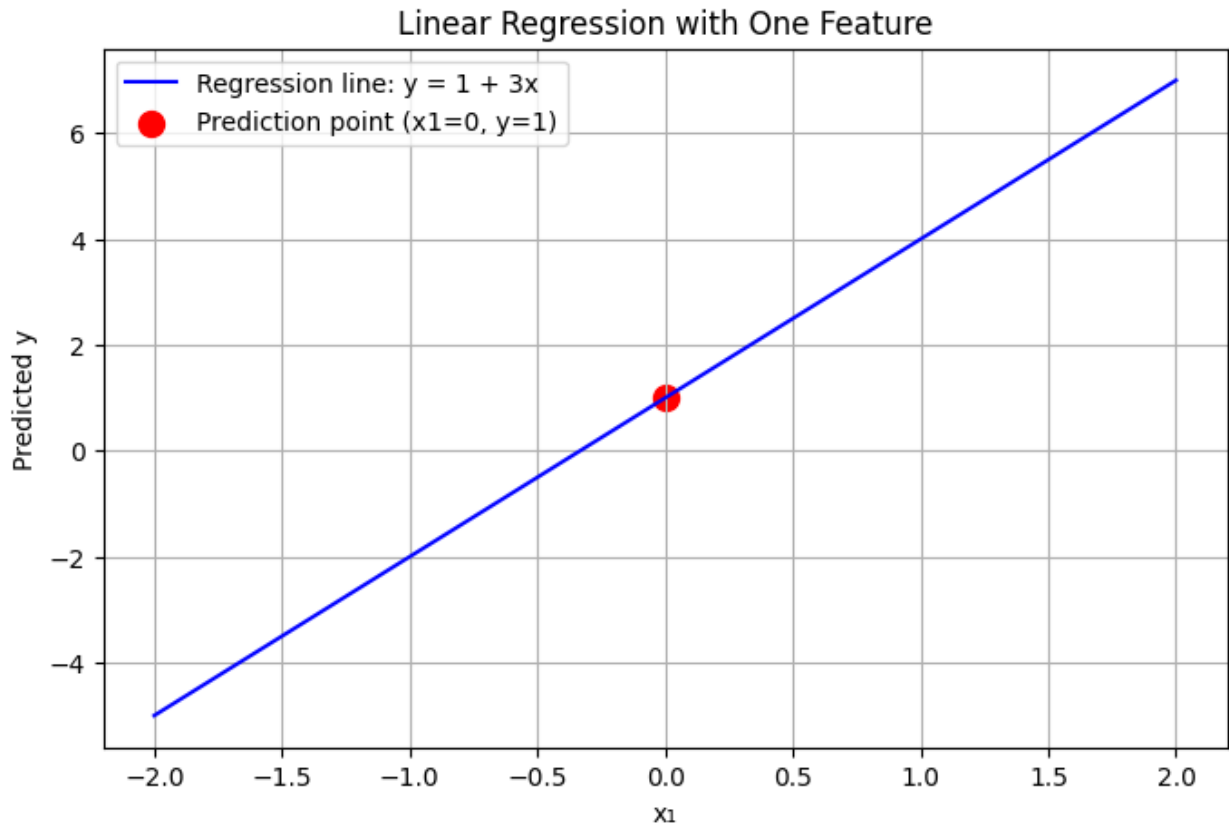
b = 1
w1 = 3
x1 = 0

y_pred = b + w1 * x1
print(f"Prediction for x1 = {x1}: {y_pred}")

x_vals = np.linspace(-2, 2, 100)
y_vals = b + w1 * x_vals

plt.figure(figsize=(8, 5))
plt.plot(x_vals, y_vals, color='blue', label='Regression line: y = 1 + 3x')
plt.scatter(x1, y_pred, color='red', s=100, label=f'Prediction point (x1={x1}, y={y_pred})')
plt.xlabel('x1')
plt.ylabel('Predicted y')
plt.title('Linear Regression with One Feature')
plt.grid(True)
plt.legend()
plt.show()
```

Prediction for x1 = 0: 1



Analysis

The prediction is calculated as $y = 1 + 3 \times 0 = 1$. The blue line in the plot represents the regression line $y = 1 + 3x$, which has a slope of 3 and an intercept of 1. When the feature value $x_1 = 0$, the product $w_1 \cdot x_1$ becomes zero, so the prediction equals the bias term. The red dot marks this exact point on the line, confirming that the prediction lies on the regression line.

Part 2

What will be the regression prediction with the following X values? Give the output. You may plot the regression line for additional points and give analysis.

Weights and Bias

$b=1$

$w_1=3$

$w_2=-2$

$w_3=4$

Additional Example #1:

$x_1=3$

$x_2=-1$

$x_3=2$

Additional Example #2:

$x_1=10$

$x_2=0$

$x_3=-10$

```
b = 1
w1 = 3
w2 = -2
w3 = 4

x1_1 = 3
x2_1 = -1
x3_1 = 2
y_pred1 = b + w1*x1_1 + w2*x2_1 + w3*x3_1
print(f"Prediction for Example 1: {y_pred1}")

x1_2 = 10
x2_2 = 0
x3_2 = -10
y_pred2 = b + w1*x1_2 + w2*x2_2 + w3*x3_2
print(f"Prediction for Example 2: {y_pred2}")

Prediction for Example 1: 20
Prediction for Example 2: -9
```

Analysis

The prediction for Example 1 is $1 + 9 + 2 + 8 = 20$. For Example 2, it is $1 + 30 + 0 - 40 = -9$. In Example 1, the negative weight w_2 and negative x_2 combine to add a positive contribution. In Example 2, the large negative term from $w_3 \cdot x_3$ (-40) outweighs the positive part. Each feature contributes independently based on its weight.